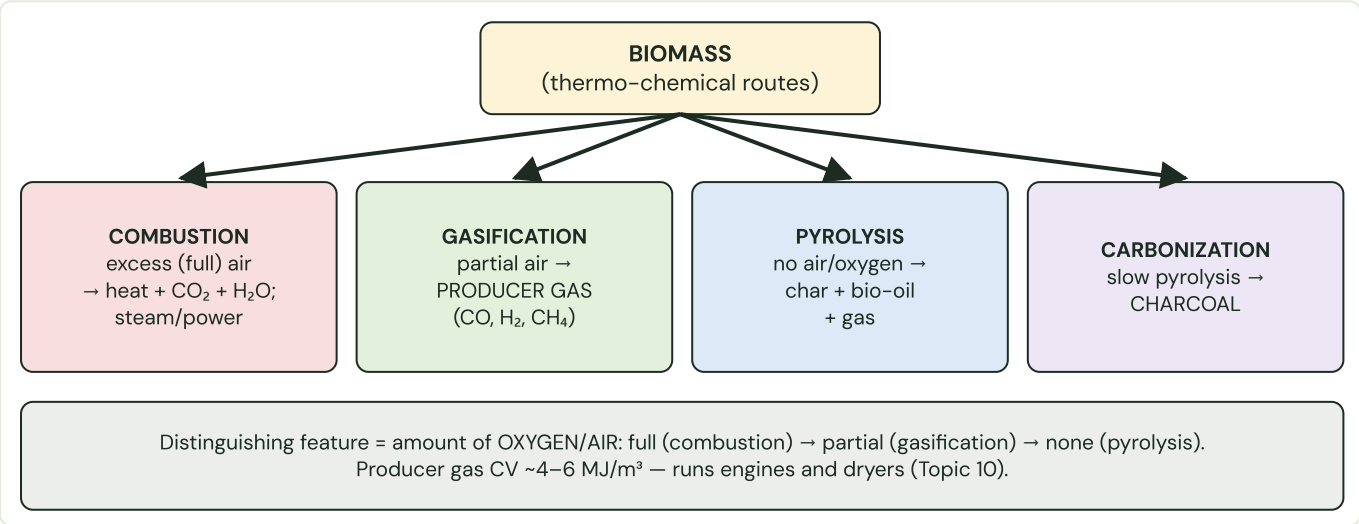
UNIT 7 · TOPIC 6

Thermo-Chemical Conversion — Combustion, Gasification and Pyrolysis

Syllabus topic: Thermo-chemical conversion of biomass: direct combustion, pyrolysis and gasification

Biomass and its conversion routes

- **Biomass** = organic matter usable as fuel — crop residues, wood, energy crops, dung, agro-industrial and municipal wastes.
- **Routes:** THERMO-CHEMICAL (combustion, gasification, pyrolysis, carbonization — this topic and Topic 7) and BIO-CHEMICAL (anaerobic digestion → biogas; fermentation → ethanol — Topics 8–9).
- Biomass is carbon-neutral in principle: the CO_2 released was recently fixed from the air.



Thermo-chemical conversion of biomass (redrawn from the source figure)

The routes differ in the amount of OXYGEN/AIR supplied: full (combustion) → partial (gasification) → none (pyrolysis).

Direct combustion

- Complete burning in EXCESS air → heat + CO_2 + H_2O. Used in stoves, furnaces and biomass power boilers (steam → turbine).
- Simple, but efficiency depends on moisture (wet biomass wastes heat as latent heat of vaporisation) and on good air–fuel mixing.
- Improved chulhas and fluidised-bed/grate combustors raise efficiency and cut smoke.

Gasification

- PARTIAL (sub-stoichiometric) combustion at high temperature converts solid biomass into a combustible **PRODUCER GAS** — chiefly CO, H_2 and CH_4, with N_2 and CO_2; calorific value **~4–6 MJ/m³** (a "low-CV" gas).
- Gasifier types: updraught, **downdraught** (clean gas for engines), fluidised-bed.

- Producer gas runs IC engines (dual-fuel/gas engines), dryers and thermal applications — an important decentralised farm-power route (Topic 10; Unit 4, Topic 1b).

Pyrolysis

- Thermal decomposition in the ABSENCE of oxygen, giving three products: solid **CHAR**, liquid **BIO-OIL** (pyrolytic oil) and combustible **GAS**.
- The proportions depend on temperature and heating rate: **FAST** pyrolysis maximises bio-oil; **SLOW** pyrolysis maximises char (Topic 7).

Updated information — bioenergy in India (as given in the source notes, July 2026)

Item	Status
National Bioenergy Programme	MNRE Phase-I, 2021-22 to 2025-26, outlay ₹858 crore — supports biomass power, gasifiers, briquette/pellet units, biogas and Bio-CNG (CBG under SATAT)
Ten-year additions	~2,361 MW biomass power; ~228–254 MWe waste-to-energy; ~2.88 lakh biogas plants
Co-firing	Biomass PELLET co-firing in coal thermal plants is mandated — uses surplus crop residue and curbs stubble burning
Non-fossil context	Part of India's ~283–289 GW non-fossil capacity (Topic 1)

Practice MCQs

Q86. The distinguishing feature among thermo-chemical routes is the amount of:

(a) Water (b) Oxygen/air supplied
(c) Pressure (d) Catalyst

Q87. Direct combustion of biomass uses:

(a) No air (b) Partial air
(c) Excess (full) air (d) Pure nitrogen

Q88. Gasification is carried out with:

(a) Excess air (b) Partial (sub-stoichiometric) air
(c) No air (d) Pure oxygen only

Q89. The main combustible components of producer gas are:

(a) CO, H_2 and CH_4 (b) CO_2 and N_2
(c) O_2 and H_2O (d) SO_2 and NO

Q90. The calorific value of producer gas is roughly:

(a) 0.5 MJ/m³ (b) 4–6 MJ/m³
(c) 38 MJ/m³ (d) 120 MJ/m³

Q91. The gasifier that gives the cleanest gas suitable for engines is the:

(a) Updraught (b) Downdraught
(c) Open fire (d) None

Q92. Pyrolysis is thermal decomposition in the:

(a) Presence of excess air (b) Absence of oxygen
(c) Presence of water only (d) Presence of catalyst only

Q93. The three products of pyrolysis are:

(a) Char, bio-oil and gas
(b) Only char
(c) Only gas
(d) Ash and water

Q94. Fast pyrolysis is optimised to maximise:

(a) Char
(b) Bio-oil (pyrolytic oil)
(c) Ash
(d) CO_2

Q105. Biomass is considered broadly carbon-neutral because:

(a) It has no carbon
(b) The CO_2 released was recently fixed from the atmosphere
(c) It absorbs SO_2
(d) It emits O_2

Answers

Q86 — (b) Oxygen/air supplied — *Routes ranked by oxygen — combustion (excess) → gasification (partial) → pyrolysis (none).*

Q87 — (c) Excess (full) air

Q88 — (b) Partial (sub-stoichiometric) air

Q89 — (a) CO, H_2 and CH_4 — *Producer gas = CO + H_2 + CH_4.*

Q90 — (b) 4–6 MJ/m³ — *CV 4–6 MJ/m³.*

Q91 — (b) Downdraught — *The downdraught gasifier gives engine-grade gas.*

Q92 — (b) Absence of oxygen

Q93 — (a) Char, bio-oil and gas — *Pyrolysis gives char + bio-oil + gas.*

Q94 — (b) Bio-oil (pyrolytic oil) — *Fast pyrolysis → bio-oil.*

Q105 — (b) The CO_2 released was recently fixed from the atmosphere

Sources: Rai — Non-Conventional Energy Sources; Sukhatme; Duffie & Beckman; MNRE Bio-Energy & Wind programmes and PIB releases (as given in the source notes, July 2026); Betz (1919) actuator-disc theory; NPTEL renewable-energy & biomass lectures.